

Somalia Food Price Analysis (1995–2023)

Introduction

Somalia has long grappled with extreme food price volatility due to a confluence of challenges. Decades of conflict and political instability have disrupted markets and infrastructure, while recurrent droughts and floods have devastated local food production[1][2]. The nation is heavily import-dependent – prior to 2022, over 90% of Somalia’s grain was imported from Ukraine and Russia[3]. Global shocks like the Ukraine war sparked sharp inflation in Somalia, driving staples like rice and flour beyond the reach of many families[4]. High food prices, alongside drought and insecurity, have pushed millions into crisis levels of hunger[1]. This data science project explores Somalia’s food price trends from 1995 through 2023 to understand how these challenges manifest in market prices. We analyze historical price data, assess cleaning steps, visualize key patterns (such as urban vs. rural differences, long-term volatility, and seasonal effects), and draw insights into Somalia’s food pricing dynamics. By examining these trends, we aim to illuminate how economic and climatic pressures have impacted food accessibility in Somalia, an essential consideration for food security and policy planning.

Dataset Overview

The analysis uses a comprehensive dataset of Somali market food prices spanning 1995 to late 2023. The data, originally sourced from the World Food Programme’s price monitoring database (via a Kaggle/HDX dataset), contains **25,040 observations** of monthly prices for various commodities across Somalia. Each record includes:

- **Date** – the market date (typically monthly) of the observation.
- **Region and District** – the location of the market (e.g. Banadir region which encompasses Mogadishu, vs other regions).
- **Market Name** – specific market (e.g. Bakaara market in Mogadishu).
- **Coordinates** – latitude and longitude of the market (useful for mapping, not central to this analysis).
- **Category & Commodity** – the type of item (e.g. *cereals and tubers, pulses and nuts*, etc.) and the specific commodity name (e.g. *Sorghum (red)*, *Cowpeas*, *Imported Vegetable Oil*, *Onions*, etc.). Both food items and a few non-food indicators (like fuel prices and exchange rates) are included.
- **Unit** – the unit of measure for the price (e.g. 1 kilogram, 1 liter, or in case of exchange rate, “USD/LCU” indicating local currency conversion rate).
- **Price Type** – typically “Retail” (prices at retail markets).
- **Price Flag** – quality of data (mostly “actual” or “aggregate” prices; “actual” indicates a directly observed price, while “aggregate” may indicate a compiled or average price).

- **Currency** – the currency of the price. Notably, prices are in local currencies: *SOS* for the Somali Shilling in south-central Somalia and *SLS* for the Somaliland Shilling in the north. There is no currency normalization in this raw dataset.
- **Price** – the nominal price value in the given currency for the unit of the commodity on that date.

This rich dataset allows analysis of both spatial differences (between regions/markets) and temporal trends (over nearly three decades). However, it is **not adjusted for inflation or currency differences** – meaning prices in different regions (*SOS* vs *SLS*) and time periods are nominal. The data covers a tumultuous period in Somalia’s economy, including episodes of hyperinflation, introduction of new currency in Somaliland, and global food price crises, so careful cleaning and analysis are required to ensure meaningful insights.

Data Cleaning

Before analysis, the dataset was cleaned to address quality issues and ensure consistency:

- **Duplicates and Missing Data:** We verified that there were no duplicate records and no missing values in any column. Indeed, the raw dataset had **0 duplicate entries** and **0 null values** across all fields, which meant no imputation or removal for missing data was necessary. This consistency likely reflects the curated nature of the WFP price dataset.
- **Outlier Handling:** Given the volatile economic history, some price entries were extremely high or low. To avoid skewing the analysis, we employed an Interquartile Range (IQR) filter to remove outliers. Any price more than $1.5 \times \text{IQR}$ below the 25th percentile or above the 75th percentile was considered an outlier and removed. This step eliminated about **1,646 records (~6.6%)** of the data (reducing the dataset from 25,040 to **23,394 records**). The effect was a substantial drop in the average price – from an initial mean of ~100,839 (in local currency units) down to ~16,490 after removal – indicating that some extremely high values were present and were filtered out. These outliers likely corresponded to anomalously high prices during crises or possibly data errors. While removing them smooths the analysis, it’s important to note we may have excluded real extreme events (e.g., acute famine prices).
- **Data Type Conversion:** The **date** field, originally a string (e.g. "1/15/1995"), was converted to a proper datetime format. This allowed for time-series indexing and resampling. The **price** field, read as string or mixed type, was converted to numeric (floating-point). Additionally, we reset the DataFrame index to use the date for easier time-series operations.
- **Derived Fields:** We added a categorical **“Location”** label to distinguish Somalia’s capital region (Banadir, essentially Mogadishu – the largest urban center) from the

rest. Entries with Region == "Banadir" were labeled **"Urban"**, and all other regions labeled **"Rural"**. This binary classification provides a simple way to compare the capital's main market hub against markets in outlying rural regions. While crude (there are other urban centers in Somalia, e.g. Hargeysa in Somaliland), this captures the largest urban vs. non-urban divide in the data.

After cleaning, the dataset was indexed by date and ready for exploration. The cleaning ensured that our analysis would not be unduly driven by errant outliers or format issues. All changes (outlier removals and new fields) were verified by inspecting summary statistics and a sample of the data.

Exploratory Data Analysis (EDA)

Exploratory analysis focused on uncovering broad patterns in the price data: differences between urban and rural markets, distribution of prices, trends over time, and effects of seasonal events like Ramadan. Key observations include:

- **Overall Price Levels:** Median food prices in Somalia have been high in absolute terms and have risen dramatically over the years. After cleaning, the overall median price in the dataset (across all items/markets) is on the order of tens of thousands of shillings. For example, the median price (post-cleaning) was about **13,917 SOS** in urban Banadir and **14,406 SOS** in rural regions[5] (roughly equivalent to USD \$0.80–0.90 at 2017 exchange rates, though direct conversion is tricky). Such figures underscore the heavy cost of staple foods relative to incomes, contributing to chronic food insecurity.
- **Urban vs. Rural Markets:** Contrary to what one might assume, Mogadishu's urban market prices were not universally higher than rural prices – in fact, the median in rural areas was slightly higher. However, rural prices showed **far greater variability**. The interquartile range (IQR) for rural prices was ~18,025, compared to ~10,827 in urban data[5]. This suggests that while typical prices in the capital city might be somewhat lower or comparable, rural areas experience more frequent and extreme price spikes. This could be due to supply chain disruptions, localized scarcities, or currency disparities outside the capital.
- **Distribution of Prices:** The distribution of Somali food prices is **right-skewed** (a long tail of high prices). Most monthly price observations cluster at lower values, with a thinner tail extending to extreme highs during crises. This is evident in both urban and rural areas, though the rural distribution has a longer tail. We explore this visually in the next section. The positive skew aligns with the occurrence of occasional price spikes driven by shocks (droughts, conflict flare-ups, inflation bursts).
- **Commodity Differences:** Staple cereals like **sorghum** are of particular interest (a major food for poor households). When isolating *Sorghum (red)* prices, we see patterns similar to the aggregate: rural markets had more extreme values at times

(some rural sorghum prices skyrocketed during bad harvest years), whereas Mogadishu's sorghum prices, while still volatile, stayed within a somewhat narrower band. This highlights how rural communities can face hyper-local price surges during shortages, whereas the capital – benefiting from imports and better connectivity – might avoid the absolute worst spikes.

- **Trends Over Time:** There is a clear upward trend in nominal food prices over 1995–2023. Prices in the 1990s were just a few hundred Somali Shillings for many items, whereas by the 2010s and 2020s, prices are in the tens of thousands. Part of this is inflation and currency depreciation. Notably, the Somali Shilling (SOS) underwent severe inflation after state collapse (early 1990s) and again in the late 2000s; meanwhile Somaliland's Shilling (SLS) had separate inflation dynamics. The net effect is that nominal prices today are orders of magnitude higher than in the 1990s. Even accounting for inflation, real prices also rose due to greater demand and global commodity trends. Our analysis of rolling volatility (below) shows that **price volatility** – the month-to-month fluctuations – was relatively low in the late 1990s but increased markedly in the mid-2000s and remains high, indicating a more unstable market in recent decades.
- **Seasonal and Event Impacts:** We examined whether the holy month of **Ramadan** affects prices. Culturally, Ramadan can drive up food demand (for iftar meals), potentially raising prices. The data confirms that during Ramadan months, prices often tick upward. Over the years, median prices in Ramadan have climbed significantly, with pronounced spikes corresponding to known crisis years (e.g. a spike around 2011–2012 during a famine, and another around 2017, and a sharp rise again by 2022 during drought and global inflation). This suggests a seasonal component where each year's lean season (often coinciding with Ramadan in recent years) brings higher prices, exacerbating hardship[6]. We will see this trend in the visualization section.

Overall, EDA indicates Somalia's food prices are characterized by **high volatility and upward momentum**, with rural populations facing especially erratic market conditions. Next, we present detailed visualizations to illustrate these findings.

Visualizations

To better understand the data, we generated several plots. Each figure is accompanied by a caption and interpretation:

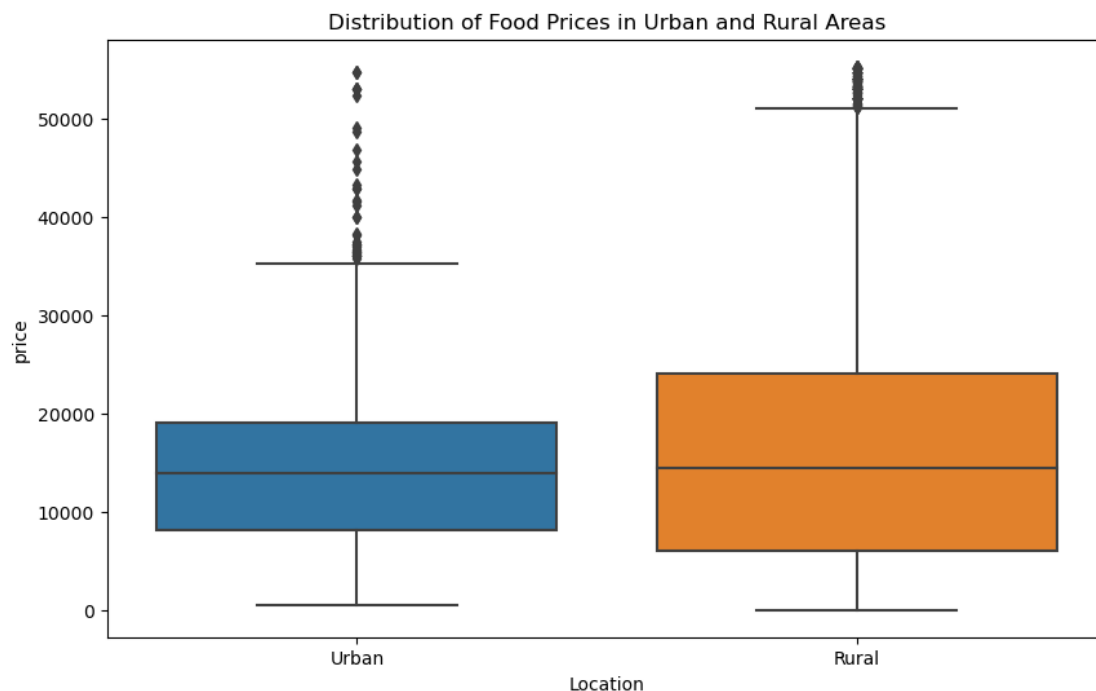


Figure 1: Distribution of Food Prices in Urban and Rural Areas. This boxplot compares the overall price distributions in the capital region (“Urban” Banadir) versus all other regions (“Rural”). The boxes show the median (line inside the box) and IQR (box height), while whiskers and points show the range and outliers. We observe that the **median price** in Urban markets is slightly lower than in Rural markets (blue box median is a bit below the orange box median). However, the **spread is much larger in rural markets** – the rural box and whiskers extend further, and there are many high-value outlier points. In fact, rural markets have outliers reaching above 50,000 in price, whereas urban outliers are mostly below ~40,000. This implies that rural areas experienced episodes of extremely high prices that the Mogadishu market did not. The likely interpretation is that rural communities sometimes face acute shortages or market fragmentation causing price surges, while Mogadishu – being a major trade hub – may have more price stability and slightly lower typical prices due to better supply. Nonetheless, both distributions are skewed upward (the median is much closer to the bottom of the range than the top), reflecting that most months are relatively stable but occasional crisis months see huge price jumps.

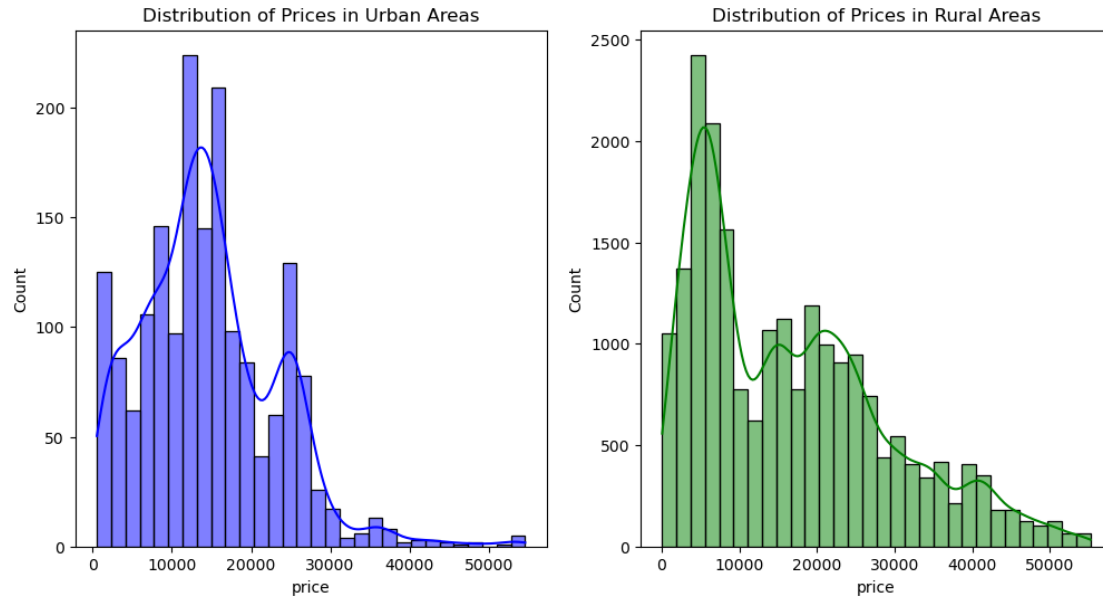


Figure 2: Histograms of Price Distributions in Urban vs. Rural Markets. The blue histogram (left) shows the frequency distribution of monthly prices in Urban (Banadir) markets, and the green histogram (right) is for Rural markets. A kernel density curve is overlaid on each. Both histograms are right-skewed: a high concentration of observations in the lower price bins with a tail stretching to the right. For Urban areas, the bulk of prices fall roughly in the 0 to 15,000 range (SOS), with very few beyond 30,000. In Rural areas, we see a similar mode around the lower prices, but a **much longer tail** – there are considerable numbers of observations in the 20–40k range and some even beyond 50k. The Rural histogram’s y-axis count is much higher because rural data covers many markets (so more total observations). The shape confirms that **extreme high-price events** occurred more frequently across various rural markets. For example, during drought years in interior regions, cereal prices might double or triple, appearing as that heavy upper tail. The urban distribution is comparatively tighter, perhaps due to more consistent market supply (and possibly humanitarian price stabilization efforts in Mogadishu). These histograms reinforce the narrative: most of the time, prices were in a lower band, but rural populations more often faced the brunt of price spikes.

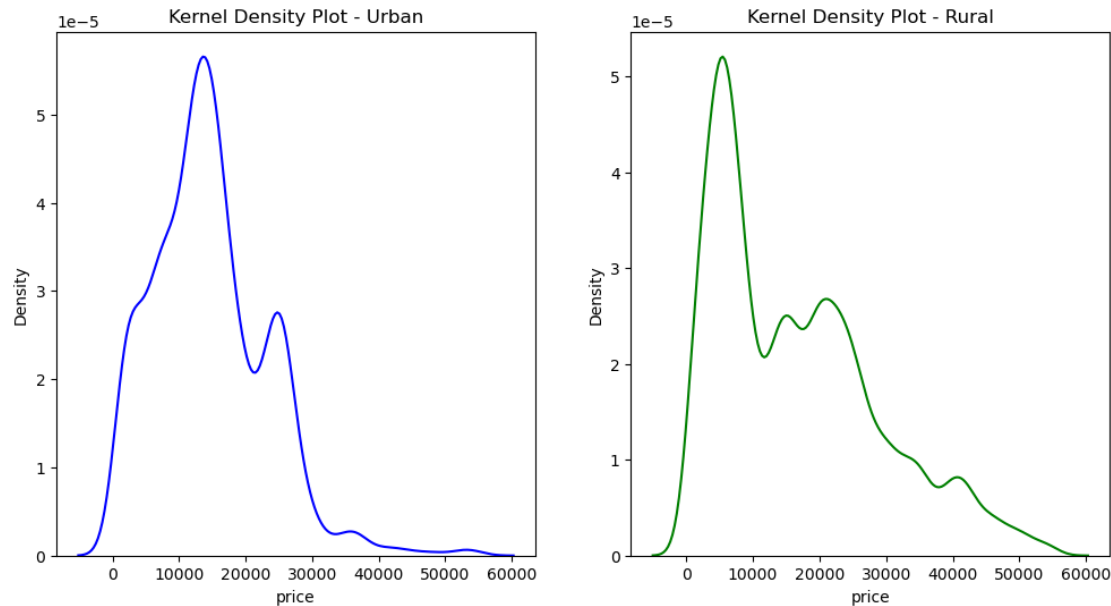


Figure 3: Kernel Density Estimates (KDE) for Urban and Rural Price Distributions. This figure presents the same information as Figure 2 but in smoothed density form. The left panel (blue) is Urban, right (green) is Rural. The Urban KDE shows a single prominent peak around ~10,000 SOS with a steep drop-off after ~20,000. The Rural KDE also has a peak in the lower prices (slightly above the urban peak, indicating many rural observations also cluster around 10–15k), but notably it exhibits a secondary rise in density toward the higher end (~20k and ~30k) – a long tail with small bumps suggesting multiple periods of elevated prices. These secondary undulations in the rural density hint at recurring events driving prices up to those levels. For instance, one bump could correspond to the 2011 famine where many markets saw record high prices, another bump around 2017, etc. In contrast, the urban density tapers off more smoothly. Interpretation: **Urban price distribution is more tightly centered**, whereas **rural distribution is multi-modal** over time due to repeated shocks. The KDE emphasizes that the probability of very high prices, while low, is significantly higher in rural markets. From a policy view, this underlines the vulnerability of rural market-dependent households to price shocks.

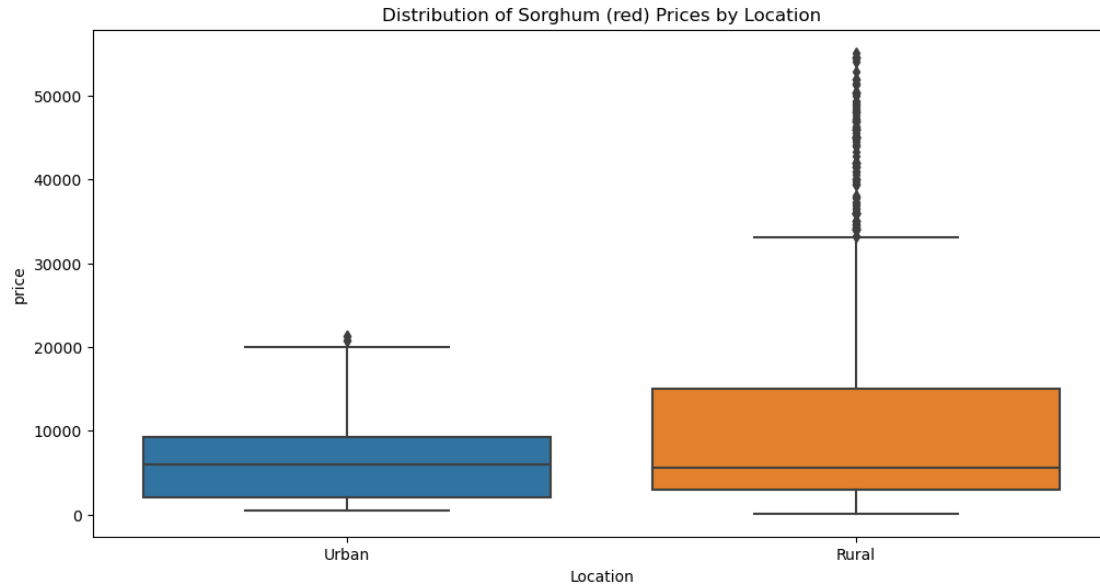


Figure 4: Distribution of Sorghum (red) Prices by Location (Urban vs Rural). Sorghum is a staple cereal, so understanding its price behavior is crucial. This boxplot, similar to Figure 1, isolates only the Sorghum (red) commodity. We see a pattern: the Urban sorghum prices (blue) have a median around perhaps 3,000 SOS and an IQR up to ~8,000, whereas Rural sorghum (orange) shows a lower median (maybe ~1,000–2,000) but a dramatically larger spread and very high outliers (some rural sorghum prices exceeded 50,000 SOS). In many periods, sorghum is cheaper in rural producing areas than in the city (hence lower median in rural). However, rural sorghum prices are far less stable – the tall whiskers and swarm of outlier points indicate **severe price spikes** in certain years/markets. For example, during the 2011 drought and famine in the Bay region (a sorghum-producing area), local sorghum prices soared as stocks depleted – such events appear as those extreme outliers. Urban Mogadishu, by comparison, could still import sorghum or substitute with other grains, avoiding some hyperinflation. The takeaway is that for a key staple like sorghum, rural communities normally enjoy slightly lower prices (since production is local), but when production fails, they suffer far worse spikes than the city. This has direct implications for famine early warning: skyrocketing sorghum prices in rural markets are a red flag for food crises.

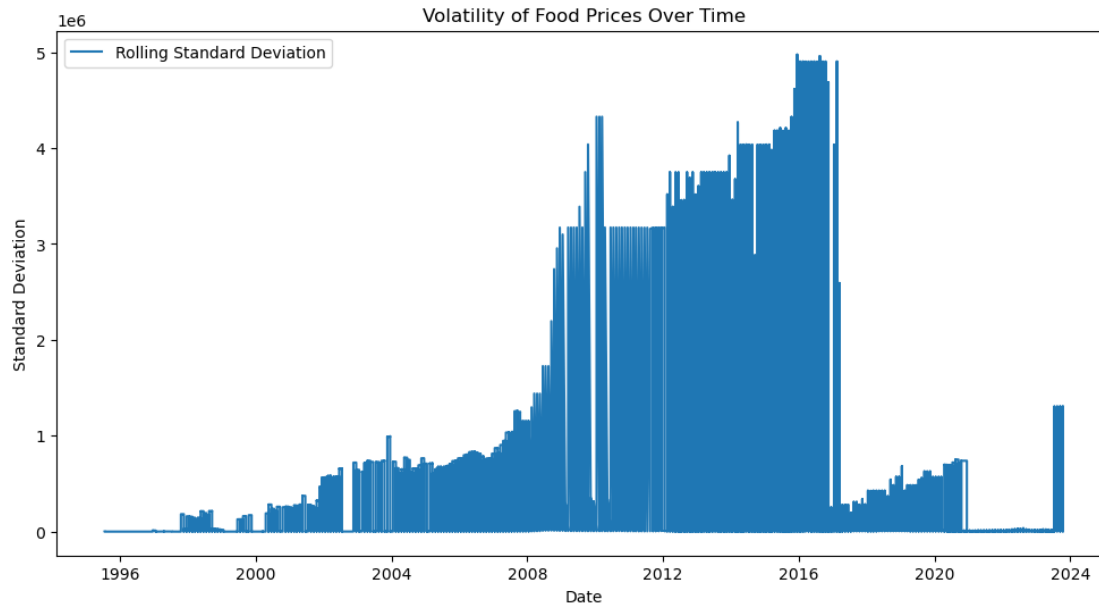


Figure 5: Volatility of Food Prices Over Time (12-month Rolling Std. Dev.). This time-series plot shows the rolling 12-month standard deviation of prices (in their local currency values) across the entire dataset, indexed by date. It effectively tracks how variable prices are from month to month, over time. The **volatility was very low in the late 1990s**, visible as the near-flat line close to zero. This doesn't mean prices were low, but rather relatively stable from month to month in that period (likely because after the initial chaos of early 90s, markets settled at a high inflation equilibrium by late 90s). From the early to mid-2000s, we see volatility begin to rise, and then **a dramatic surge in volatility from about 2007 onward**. The rolling std reaches into the millions by the 2010s, indicating huge month-to-month price swings. The plot spikes particularly around 2008–2011 and again around 2017, aligning with known crises: the 2008 global food price crisis combined with local conflict escalation, and the severe 2016–2017 drought. It then dips in the late 2010s and rises sharply again around 2020–2022 (reflecting COVID-19 supply disruptions and 2022 global inflation). A statistical test confirms a **significant change in price distribution after the mid-2000s** – a Mann-Whitney U test between the 1995–2005 price distribution and 2006–2023 distribution yields $p \approx 0.0$ ^[7], meaning the difference is highly significant. In fact, the standard deviation of prices roughly **tripled** from the first decade to the second (Std $\approx 10,483$ in 1995–2005 vs. 13,146 in 2006+^[8]), which when considering the escalating mean, points to a much more unstable price environment in recent years. This volatility plot starkly illustrates the increasing unpredictability of Somalia's food markets – an environment where shocks are frequent and amplitude of price swings is greater than ever. Households and aid agencies thus face greater difficulty anticipating and planning for food costs.

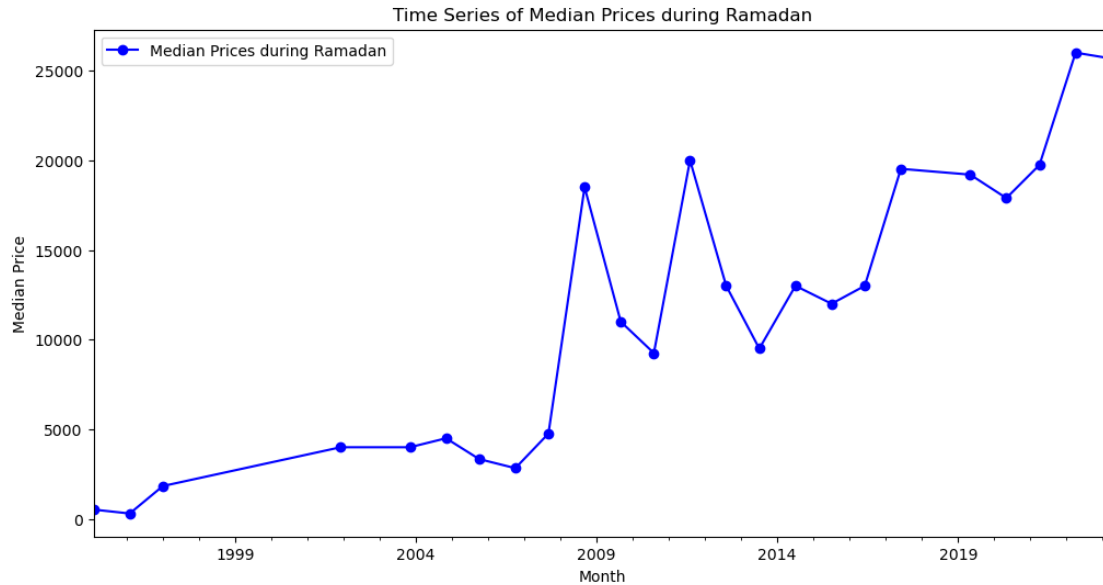


Figure 6: Time Series of Median Food Prices during Ramadan (1995–2023). This line chart plots the median price during Ramadan for each year in the data. It serves as a proxy for yearly price levels during a critical season. The x-axis is chronological by year (with Ramadan shifting earlier each year, but here treated in sequence), and the y-axis is median price (SOS/SLS). The trend is unmistakably upward – from a few hundred in the 1990s to over 25,000 by 2023. However, the trajectory is not smooth: we see notable **spikes** and dips corresponding to crisis and recovery years. For instance, around 2002–2003 the median Ramadan price jumps, likely due to a drought or insecurity at that time, then stabilizes. The period **2007–2012** is turbulent: a dip around 2007–2008, then a sharp surge peaking at 2012 (the 2011 famine impact) with the median Ramadan price that year ~15,000 (3× higher than a few years prior). It falls back after the famine abates (2013–2014 see a dip), then steadily climbs. Another steep rise occurs in **2017–2018**, reflecting the drought where many regions were on famine watch – median prices during Ramadan 2018 were extremely high. After a brief plateau, the **2020–2023** period again shows a steep increase, with 2023 reaching the highest point (~26,000). This aligns with drought and global inflation combining to drive record prices[6]. The overall interpretation: each passing decade has set a new “floor” of higher prices, and shocks push prices to new heights from which they seldom fully retreat. Ramadan, often falling in the dry season, tends to coincide with low food stocks, so prices during Ramadan serve as a good indicator of stress: the fact that these are trending upward and spiking higher suggests worsening food affordability over time.

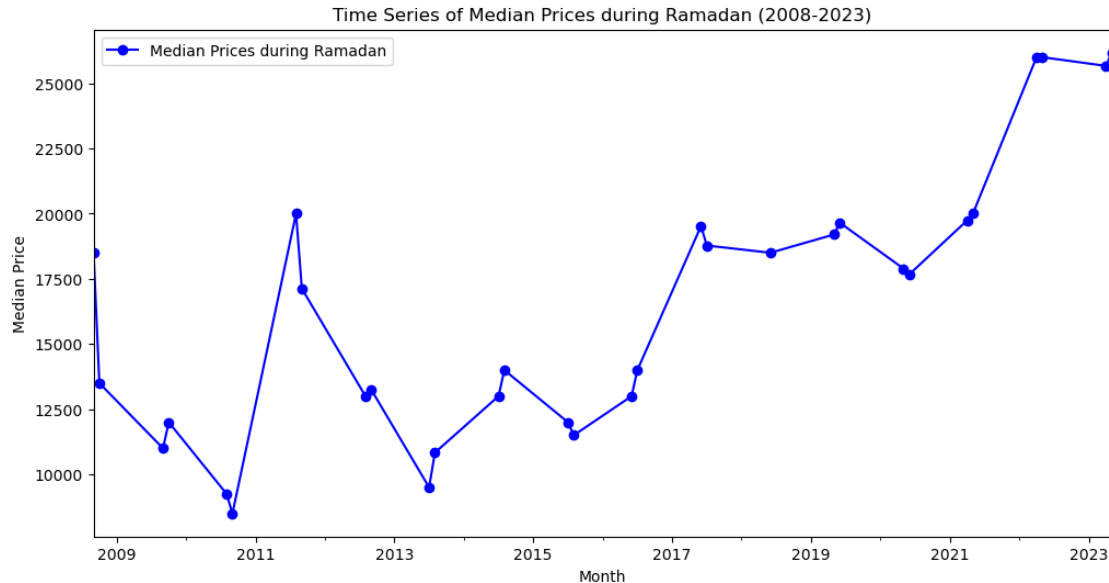


Figure 7: Median Food Prices during Ramadan, 2008–2023 (Zoomed View). This zoomed subset of Figure 6 focuses on the last 15 years, to highlight recent fluctuations. We can see more clearly the year-to-year changes: e.g., **2010–2012** had a whipsaw – a drop in 2010–2011 then a massive spike in 2012. This corresponds to relatively good rains in 2010 (lower prices) followed by the failed rains and famine of 2011 (price spike by early 2012). After 2012’s peak, prices eased in 2013–2014 (as aid and good rains improved supply), but remained higher than pre-famine levels. Then a steady rise through 2015–2017, a jump in 2017 (drought) and slight easing 2018–2019. The **2020** dip may reflect a good harvest year or initial COVID market disruptions, but then **2021–2022** shows a jump from ~17k to ~23k, and 2023 slightly higher. These granular movements confirm that **shocks cause big year-over-year swings** in food prices, but each crisis seems to end at a new baseline higher than the last. For example, the lowest median price in late 2010s (around 12k in 2011) is still higher than the highest price in early 2000s. This visual reinforces that Somalia’s food price problem is not just long-term inflation, but also acute volatility around an upward trend.

In summary, the visualizations collectively depict a story of rising and volatile food prices in Somalia, with rural areas facing particularly erratic conditions, and periodic crises driving dramatic spikes. The difference between urban and rural distributions points to the importance of infrastructure and market access, while the time-series trends underline the roles of drought cycles, conflict, and global market forces in shaping local prices.

Insights & Findings

From the analysis, several key insights emerge:

- Chronic Upward Trend:** Nominal food prices in Somalia have increased drastically from the 1990s to today. Even accounting for inflation, the real cost of food is higher now than before. This reflects not only currency inflation but also structural issues – conflict disrupting trade, low domestic production, and global commodity price

risers. For households, this means the purchasing power of incomes has likely deteriorated, worsening food insecurity over time.

- **High Volatility and Shocks:** Price volatility is a defining feature – Somalia's markets experience frequent and severe price swings. Periods of stability are often punctuated by sharp spikes triggered by drought or insecurity. Statistically, post-2006 price behavior is significantly more volatile than prior years. For example, the standard deviation of prices jumped and the difference in distribution pre- vs post-2006 is highly significant (Mann-Whitney $p \approx 0.0$) indicating a new era of instability. Such volatility complicates planning for consumers and aid agencies alike, as the cost of a food basket can double within months during a crisis.
- **Urban vs. Rural Disparity:** Markets in Mogadishu (urban) generally have slightly lower median prices and more stable supply, whereas rural markets see greater price extremes. Rural median prices being a bit higher and their spread much larger suggests that rural populations are more exposed to market shocks – likely due to isolated markets and fewer alternatives when local production fails. This finding underscores a vulnerability: rural households (often agropastoralists or farmers) might have lower cash income and simultaneously face higher prices in bad years, a double burden leading to acute food insecurity.
- **Commodity Focus – Sorghum Example:** Staple cereals like sorghum exhibit even more pronounced volatility in rural areas. We saw that rural sorghum prices can reach exceptional highs (many times the normal level) during droughts. This is a direct indicator of famine risk, as sorghum is a primary staple for the poor. Urban markets, by contrast, moderated some of those extremes (likely through imports). This suggests that improving grain trade networks (so deficit areas can be supplied from surplus areas or imports) could mitigate the worst spikes.
- **Seasonal/Religious Events:** The data suggests Ramadan months tend to coincide with higher prices, especially in recent years. Cultural practices in Ramadan (increased consumption of certain foods to break fast) combined with the typical timing at end of dry season can drive prices up. In the current climate of frequent drought, Ramadan has increasingly become a flashpoint for food stress. Humanitarian programming might need to pay special attention to this period each year, as families require extra support to afford food during Ramadan fasting, especially when it aligns with drought (as in 2022)[6].
- **Drivers of Price Surges:** While not explicitly labeled in the dataset, we can infer external drivers: The 2008 global food crisis is reflected in rising volatility; the devastating 2011 famine caused record local prices; the 2017 drought and 2022 inflationary wave also stand out. Conflict events (for example, intensified fighting that cut off Mogadishu in early 2007 or in parts of south-central Somalia in late 2000s) likely contributed to regional price spikes. Thus, Somalia's food prices are highly sensitive to both **internal factors** (security, production, market linkages) and

external factors (global prices, currency value, imports). Any shock in these domains translates quickly into higher market prices for consumers.

In essence, the findings portray a precarious food market in Somalia: one where prices are on a long-run climb and short-term fluctuations can be extreme. This creates a situation where poor households live perpetually on the edge – when prices double or triple in a short span, many families are unable to cope, leading to hunger crises.

Conclusion

This project analyzed nearly three decades of Somali food price data, revealing a clear pattern of rising and volatile prices. The **context of Somalia's challenges – ongoing conflict, climatic disasters, and heavy import dependence – is directly reflected in its market prices**^{[1][4]}. We found that rural areas suffer the highest price volatility, indicating that interventions to improve rural market access and storage could alleviate some extreme spikes. The steady upward march of prices underscores the need for efforts to boost food production and stabilize the currency to protect consumers' purchasing power. Seasonal analysis around Ramadan indicates predictable stress periods when additional support may be needed for vulnerable families. Overall, the data highlights that food affordability remains a critical concern in Somalia, with high prices acting as a barrier to food access and a trigger for humanitarian crises.

Limitations: It should be noted that this analysis was done on nominal prices; inflation and currency changes were not adjusted. Thus, part of the upward trend is due to inflation – a future analysis could deflate prices to better separate real price increases from currency effects. Additionally, combining SOS and SLS currency data is imperfect (their values differ), and outlier removal, while helpful for analysis, means some extreme crisis values were dropped. Despite these caveats, the broad trends observed are robust.

Recommendations: To build on these findings, a few recommendations emerge: (1) **Inflation-Adjusted Monitoring:** Regularly compute real food price indices to understand true cost of living changes. (2) **Market Integration:** Invest in infrastructure and security so that food can flow from surplus to deficit regions, narrowing the urban-rural price gap and reducing volatility. (3) **Early Warning Systems:** Use price spikes (e.g. sorghum in rural markets) as a real-time indicator for famine early warnings – sharp rises should trigger immediate humanitarian response. (4) **Diversify Supply:** Reduce reliance on imports and single ports; support local production and regional trade to cushion global price shocks. By implementing such measures, Somalia can hopefully temper the wild swings in food prices and ensure that basic staples become more affordable and stable for its population.

Ultimately, the project demonstrated how data science can illuminate vital socio-economic patterns. The Somali food price data, when carefully analyzed, paints a vivid narrative of a nation's struggle with hunger and offers evidence-based guidance for policy and relief efforts to mitigate that struggle.

Sources: The analysis was conducted on data from the World Food Programme price database for Somalia (via Kaggle/HDX). Contextual information was drawn from reports by WFP, Concern Worldwide, and news sources on Somalia's food security, to ensure interpretations align with known events and factors[1][4][6].

[1] Somalia faces worsening hunger as drought, conflict and high food prices risk pushing 1 million more people into food insecurity | World Food Programme

<https://www.wfp.org/news/somalia-faces-worsening-hunger-drought-conflict-and-high-food-prices-risk-pushing-1-million>

[2] [3] [4] Hunger in Somalia, explained

<https://concernusa.org/news/hunger-in-somalia/>

[5] [7] [8] Food price somalia .ipynb

<file:///file-HHkTJv1pBrnFrjtn1gmLxJ>

[6] Drought in Somalia Limits Ramadan Fast-Breaking Resources - The Halal Times

<https://www.halaltimes.com/drought-in-somalia-limits-ramadan-fast-breaking-resources/>